

Functional Annotation Report: dapA (Q88JL0) — 4-hydroxy-tetrahydrodipicolinate synthase in *Pseudomonas putida* KT2440

Target: dapA / dapA-II (OrderedLocusName PP_2639), UniProt Q88JL0 **Organism:** *Pseudomonas putida* (strain ATCC 47054 / DSM 6125 / KT2440) (PSEPK) **Family:** DapA family; Class I aldolase; (β/α)₈ TIM-barrel (Aldolase_TIM IPR013785; DapA IPR005263; DHDPS PF00701; Schiff-base aldolase signature IPR020625) **EC:** 4.3.3.7 (formerly 4.2.1.52)

1. Summary (Answer to the Research Question)

dapA (Q88JL0) encodes 4-hydroxy-tetrahydrodipicolinate synthase (HTPA synthase), historically called dihydrodipicolinate synthase (DHDPS). Its primary function is to catalyze the first committed, rate-limiting step of the **diaminopimelate (DAP) pathway** to L-lysine: the aldol-type condensation of **pyruvate** with **L-aspartate-4-semialdehyde (ASA)** to give **(2S,4S)-4-hydroxy-2,3,4,5-tetrahydrodipicolinate (HTPA)** (Rhea RHEA:34171), a cyclic product that spontaneously dehydrates to dihydrodipicolinate. The enzyme is a **soluble, cytoplasmic homotetramer** (dimer-of-dimers) that operates through a **Class I aldolase, ping-pong mechanism** involving a Schiff-base intermediate on a conserved active-site lysine, and is subject to **allosteric feedback inhibition by L-lysine**. The pathway supplies both L-lysine (protein synthesis) and meso-diaminopimelate (peptidoglycan cross-linking). In *P. putida* the gene is a **paralog ("dapA-II")** of a small DHDPS isoenzyme family, paralleling the redundant DHDPS isoforms characterized in *P. aeruginosa*.

2. Gene/Protein Identity Verification

The gene symbol **dapA** is fully consistent with the UniProt annotation and with the domain architecture: - The protein description (4-hydroxy-tetrahydrodipicolinate synthase, EC 4.3.3.7, DapA family) matches the canonical DHDPS/HTPA synthase enzyme. - Domains **PF00701 (DHDPS)**, IPR005263 (DapA), IPR002220 (DapA-like), IPR013785 (aldolase TIM-barrel) and IPR020625 (Schiff-base-form aldolase active site) are the defining features of DHDPS. - The synonym **dapA-II** (from EMBL AAN68247.1) indicates this is one of multiple dapA paralogs in *P. putida* KT2440 — a pattern experimentally documented in the related species *P. aeruginosa*.

No conflicting identity was found; literature on "DHDPS/DapA" corresponds directly to this protein family. Direct experimental characterization of the specific *P. putida* KT2440 protein Q88JL0 is limited, so its function is established by (i) strong sequence/domain homology to biochemically characterized bacterial DHDPS enzymes and (ii) direct studies in the closely related *Pseudomonas aeruginosa*.

3. Primary Function: The Reaction Catalyzed

Reaction (authoritative, UniProt/Rhea RHEA:34171 for Q88JL0 specifically): L-aspartate 4-semialdehyde + pyruvate → **(2S,4S)-4-hydroxy-2,3,4,5-tetrahydrodipicolinate (HTPA)** + H₂O + H⁺.

- DapA "converts pyruvate and L-aspartate-4-semialdehyde to 4-hydroxy-2,3,4,5-tetrahydrodipicolinic acid" and "is involved in the production of lysine and precursor molecules for peptidoglycan synthesis" (Schmitz et al., 2020,

P 32356521

- It "catalyzes the first committed step in the lysine biosynthesis pathway" (Gupta et al., 2018,

P 29744651

Sridharan et al., 2014,

P 24996798

and this condensation is the **rate-limiting step** of the DAP pathway (Impey et al., 2020,

P 31330085

- **Substrate specificity:** the two substrates are pyruvate (donor, forming the Schiff base) and ASA (acceptor aldehyde). Reported K_m values across bacterial orthologs are ~0.15–0.57 mM (pyruvate) and ~0.04–0.55 mM (ASA) (Laber et al., 1992,

P 1463470

Dogovski et al., 2013,

P 24349508

The enzyme is specific for the (S)-stereochemistry of ASA and for pyruvate as the ketoacid donor.

- **EC reclassification:** originally annotated as a dehydratase (EC 4.2.1.52) on the assumption the product was dihydrodipicolinate; once HTPA was recognized as the true, direct enzymatic product (dehydration being a subsequent, largely spontaneous step), the enzyme was reclassified as a lyase, **EC 4.3.3.7** (reflected in current annotations, e.g., Gupta et al.,

P 29744651

Gupta et al., 2018 *Vibrio*,

P 29337198

4. Catalytic Mechanism

DHDPS is a **Class I (Schiff-base) aldolase** operating by a **ping-pong bi-bi** mechanism: 1. Pyruvate binds first and forms a **Schiff base (iminium)** with the ϵ -amino group of the **absolutely conserved active-site lysine** (Lys161 in *E. coli* numbering), captured by the IPR020625 Schiff-base aldolase signature present in Q88JL0. 2. ASA then binds; aldol addition, transamination/cyclization and dehydration yield HTPA (Laber et al., 1992,

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Mutagenesis quantifies the lysine's role: *E. coli* **K161A/K161R** retain fold and substrate binding but lose ~280–750-fold in k_{cat} (0.06–0.16 s⁻¹ vs 45 s⁻¹ wild type), with essentially unchanged K_m and lysine K_i (Soares da Costa et al., 2010,

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A conserved catalytic Tyr/Thr proton-relay, contributed across the tight dimer, completes the active site.

5. Structure, Oligomeric State, and Localization

- **Fold:** each monomer is a (β/α)8 **TIM-barrel** (IPR013785) with a C-terminal α -helical domain.
- **Quaternary structure:** a **homotetramer ("dimer of dimers")**; two monomers jointly form each active site at the **tight-dimer interface**, and dimers associate around a solvent-filled central cavity (Soares da Costa et al., 2018,

P 29204887

Reboul et al., 2012,

P 22685390

Bacterial DHDPS adopts a "**head-to-head**" tetramer (contrasting the "back-to-back" plant form).

- **Functional role of oligomerization:** tetramerization rigidifies the active-site dynamics; disrupting the tetramer increases flexibility and lowers catalytic efficiency and substrate specificity (Reboul et al., 2012,

P 22685390

Atkinson et al., 2018,

P 29804823

- **Localization:** a soluble **cytoplasmic** enzyme of amino-acid/cell-envelope precursor metabolism — it acts in the cytosol, upstream of membrane-associated peptidoglycan assembly. (No signal peptide/membrane domain is present; consistent with all characterized DHDPS enzymes being cytosolic.)

5b. Protein-Specific Sequence Evidence (this study)

To move beyond family-level homology, I aligned the actual Q88JL0 sequence (288 aa) to biochemically characterized *E. coli* DHDPS (P0A6L2) by global Needleman–Wunsch (BLOSUM62). Overall identity is **33.1%** (94/284), typical for cross-genus DHDPS orthologs, but **every catalytic residue is strictly conserved and co-numbered:**

Function	<i>E. coli</i> residue	Q88JL0 residue
Schiff-base lysine (covalent catalysis)	Lys161	Lys161 ✓
Catalytic proton-relay tyrosine	Tyr133	Tyr133 ✓
Proton-relay tyrosine (2nd)	Tyr107	Tyr107 ✓
Substrate-binding threonine	Thr44	Thr45 ✓
ASA carboxylate-binding arginine	Arg138	Arg138 ✓
Allosteric lysine-binding site	His56	Gln57 ✗ (non-canonical)

Interpretation: the complete, conserved catalytic tetrad (Thr, Tyr, Tyr, Lys) plus the ASA-anchoring Arg confirm Q88JL0 is a **catalytically competent HTPA synthase** performing the canonical pyruvate + ASA condensation. However, the allosteric position 56 — where a **His or Glu is required for strong lysine feedback inhibition** (Soares da Costa et al., 2018,

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— is a **glutamine (Gln57)** in Q88JL0. This predicts that the dapA-II isoform is only **weakly, or not, lysine-feedback-inhibited**, providing a concrete molecular rationale for the reported divergence of allosteric regulation among *Pseudomonas* DHDPS paralogs (Impey et al., 2020,

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6. Regulation

DHDPS is the principal **feedback-regulated valve** of the DAP/lysine pathway. The end-product **L-lysine binds allosterically** at the tight-dimer interface (~50 Å from the catalytic sites) to inhibit activity (Skovpen et al., 2016,

P 26836694

Sensitivity is governed largely by a single residue: strong allosteric inhibition requires a **His or Glu at position 56** (*E. coli* numbering) rather than a basic residue; consequently many bacterial DHDPS enzymes are only weakly, or not, lysine-inhibited (Soares da Costa et al., 2018,

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Vauterin et al., 2000,

P 10758475

In *Pseudomonas*, distinct DHDPS isoforms differ specifically in their allosteric lysine regulation (Impey et al., 2020,

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7. Pathway Context and Biological Role

DapA opens the **diaminopimelate (DAP) pathway**, which branches from aspartate metabolism (aspartate → aspartyl-phosphate → ASA). HTPA is reduced by DHDPR (dapB) and processed through subsequent Dap enzymes (e.g., DapD/DapC/DapE or alternative routes) to **meso-2,6-diaminopimelate**, which is (i) a direct **cross-linking residue in Gram-negative peptidoglycan** and (ii) decarboxylated (LysA) to **L-lysine** for protein synthesis (Fazius et al., 2013,

P 23504110

Schnell et al., 2012,

P 22359568

Because animals lack this pathway, DHDPS is a long-standing **antibiotic/herbicide target** — though target validation must account for isoform redundancy (below).

KEGG pathway assignment (this study): PP_2639 is mapped to KEGG ortholog **K01714** (HTPA synthase, EC 4.3.3.7) and to pathway maps **ppu00300 (Lysine biosynthesis)** and ppu00261 (Monobactam biosynthesis), and to modules **M00016 (succinyl-DAP pathway, aspartate ⇒ lysine)** and **M00527 (DAP-aminotransferase pathway, aspartate ⇒ lysine)** — i.e. the enzyme feeds both DAP-pathway variants used by *P. putida*.

Genomic context (this study): dapA-II (PP_2639; genome coords 3,023,319–3,024,185, 867 bp) is **not embedded in a canonical *dap*/lysine operon**. It lies on the same strand immediately (338 bp) downstream of a **cellulose-synthase (*bcs*) operon** (PP_2634–PP_2638: cellulose synthase catalytic/translocator subunits, endo-1,4-D-glucanase, cellulose-synthase operon C protein), with a convergent GNAT acetyltransferase (PP_2640), an Fe–S protein (PP_2641) and a GntR-family regulator (PP_2642) downstream. This atypical location (versus the scattered

classical *dap* genes, and versus *dapA-I/PP_1237* at a separate locus) raises the hypothesis that *dapA-II* expression may be linked to the exopolysaccharide/biofilm-matrix context, though co-transcription with the *bcs* operon is not established.

8. *Pseudomonas*-Specific Considerations (*dapA-II*)

Unlike most bacteria, which carry a single *dapA*, *Pseudomonas* encode **multiple functional DHDPS isoforms**: - *P. aeruginosa* encodes **two functional DHDPS enzymes (PaDHDPS1, PaDHDPS2)** with similar catalytic activity but **different allosteric regulation** (Impey et al., 2020,

).

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- Deletion of the primary *dapA* in *P. aeruginosa* is **viable** (including growth in a mouse lung infection model) because homologues (e.g., PA0223, PA4188) **substitute for DapA** (Schnell et al., 2012,

).

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The UniProt synonym **dapA-II** for PP_2639 explicitly marks Q88JL0 as a **second dapA paralog** in *P. putida* KT2440, consistent with this multi-isoform arrangement. Functionally it performs the canonical HTPA-synthase reaction; regulatory sensitivity and flux contribution may be partitioned across the paralogs, and loss of a single paralog can be buffered by the others.

8.1 The *P. putida* KT2440 DHDPS/DapA paralog landscape (this study)

Querying the KT2440 proteome for PF00701 (DapA/DHDPS-like) proteins and aligning each to *E. coli* DHDPS resolves the isoform relationships:

Locus / UniProt	Gene	Length	%id vs <i>E. coli</i> DHDPS	Catalytic residues (Thr44/ Tyr107/ Tyr133/ Arg138/ Lys161)	Allosteric pos. 56	Assessment
PP_2639 / Q88JL0	dapA-II (target)	288	33.1%	all conserved	Gln (non-canonical)	competent DHDPS, likely lysine- insensitive
PP_1237 / Q88NH2	dapA-I	295	57.4%	all conserved	His (canonical)	canonical, lysine- regulated DHDPS
PP_2036 / Q88L99	—	295	26.3%	Thr44→Ser, Arg138→Thr altered	Trp	auto-annotated HTPA synthase; DHDPS activity uncertain
PP_3599 / Q88GW8	—	303	—	—	—	5-dehydro-4-deoxyglucarate dehydratase (EC 4.2.1.41) — different function
PP_1257 / Q88NF4	—	315	—	—	—	1-pyrroline-4-hydroxy-2-carboxylate deaminase (EC 3.5.4.22) — different function

Key inferences: - **dapA-I (PP_1237)** is the canonical, His56-bearing, lysine-feedback-**sensitive** "housekeeping" DHDPS. - **dapA-II (PP_2639/Q88JL0)** — the target — is a **catalytically intact but divergent paralog** (only 30.4% identical to dapA-I) predicted to be **lysine-feedback-insensitive** (Gln at position 56). This division of labor mirrors the two-isoform DHDPS system experimentally characterized in *P. aeruginosa* (Impey et al., 2020,

).

- The PF00701 superfamily in *P. putida* also contains **non-DHDPS enzymes** (a glucarate dehydratase, a pyrroline deaminase) and a residue-degenerate member (PP_2036), reinforcing that DHDPS function must be assigned from active-site conservation — not family membership alone (Impey et al., 2020,).

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9. Supported and Refuted Hypotheses

Supported: - H1 — Q88JL0 is a DHDPS/HTPA synthase catalyzing pyruvate + ASA → HTPA (domain/family homology + direct *Pseudomonas* studies). **Supported.** - H2 — Mechanism is a Schiff-base (Class I aldolase) ping-pong reaction on a conserved active-site Lys. **Supported** (mutagenesis, kinetics). - H3 — The functional unit is a cytoplasmic homotetramer with shared inter-subunit active sites. **Supported** (crystallography/AUC/SAXS across many orthologs). - H4 — Activity is feedback-regulated by L-lysine, with sensitivity set by the position-56 residue. **Supported.** - H5 — In *Pseudomonas*, dapA is one of several redundant isoforms (dapA-II). **Supported** (*P. aeruginosa* data + UniProt synonym). - H6 — Q88JL0 itself has an intact catalytic apparatus. **Supported** by direct sequence alignment: all catalytic residues (Lys161, Tyr133, Tyr107, Thr45, Arg138) are strictly conserved vs *E. coli* DHDPS. - H7 — Q88JL0 is likely lysine-feedback-insensitive (or weakly sensitive). **Supported bioinformatically** — the allosteric His56 is replaced by Gln57; requires biochemical confirmation.

Refuted / revised: - The historical belief that the direct product is dihydrodipicolinate (and that the enzyme is a dehydratase, EC 4.2.1.52) is **refuted**; the direct product is HTPA and the enzyme is a lyase (EC 4.3.3.7).

10. Limitations and Future Directions

- **No direct biochemical study of the specific protein Q88JL0** was found; function is inferred from (i) strong homology and conserved catalytic residues verified here by sequence alignment, and (ii) experimental work on orthologs/paralogs (notably *P. aeruginosa* and *E. coli*). Confirmation would benefit from recombinant *P. putida* dapA-II kinetics, lysine-inhibition assays, and a crystal structure.

- The predicted **lysine-insensitivity of Q88JL0** (Gln57 at the allosteric site, versus His/Glu in feedback-sensitive DHDPS) is a bioinformatic inference that should be tested by direct lysine-inhibition kinetics. The **division of labor** between *P. putida* DHDPS paralogs (dapA-I vs dapA-II) also remains to be characterized experimentally.
 - Whether both *P. putida* paralogs are simultaneously essential, or mutually redundant as in *P. aeruginosa*, should be tested by single and double knockouts.
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Key References (PMIDs)

32356521; 29744651; 24996798; 1463470; 20353808; 29204887; 22685390; 29804823; 26836694; 10758475; 31330085; 31943170; 22359568; 23504110; 24349508; 29337198.

Bioinformatic analyses (sequence retrieval from UniProt; Needleman–Wunsch/BLOSUM62 alignments; KT2440 paralog survey) were performed in this study.
